



DG INTERNS HUB – CYBERSECURITY INTERNSHIP

**WEEK 3 ASSIGNMENT REPORT: ADVANCED CYBERSECURITY
PRACTICAL TASK**

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INTRODUCTION

This week of my Cybersecurity Internship at DG Interns Hub focused on advanced practical security activities involving network discovery, port and service assessment, packet-level traffic analysis, vulnerability assessment, and security hardening. The main objective was to understand how a security analyst can systematically examine a network, identify active hosts and exposed services, analyze network traffic, recognize potential security weaknesses, and apply appropriate security controls in an authorized and isolated laboratory environment.

The practical assessment began with identifying the available network interfaces, IP addresses, subnet masks, and default gateway using the `ipconfig` command. A dedicated lab environment was used to perform the security assessment, including an analyzer workstation, a target gateway/server, and an isolated VirtualBox host-only network. After establishing the network configuration, Nmap was used for host discovery through a ping sweep to identify active systems within the lab subnet. This provided an initial view of the available hosts before performing more detailed security analysis.

The next stage involved advanced Nmap scanning, where SYN stealth scanning and service-version detection were performed against the authorized target. This helped identify open, closed, and filtered TCP ports and determine the services and versions running on the target system. The assessment identified services including FTP, SSH, DNS, HTTP, and HTTPS, allowing their exposure and potential security implications to be examined. Particular attention was given to services that could increase the attack surface, such as unencrypted FTP and exposed management services.

The internship task also introduced practical packet-level analysis using Wireshark. Network traffic generated during Nmap scanning was captured and examined to understand how scanning activity appears at the packet level. TCP flags such as SYN, SYN/ACK, and RST were analyzed to understand the behavior of open and closed ports. Nmap scan results were correlated with Wireshark captures, providing a clearer understanding of how network reconnaissance is reflected in actual network packets.

Following network discovery and traffic analysis, a vulnerability assessment was performed based on the services and traffic observed during the practical exercises. Identified findings were evaluated according to their potential security impact, including risks associated with plaintext communication, outdated services, and unnecessary exposed ports. The assessment demonstrated how exposed services can increase the attack

TASK 7 – Network Discovery & Basic Nmap Scanning

Objective

The objective of this task is to understand how a security analyst discovers devices within an authorized lab network and identifies exposed network services. The practical work involves examining the network configuration, identifying active hosts using Nmap host discovery, and performing a basic scan to identify open ports and services.

Lab Environment

The assessment was performed in an authorized laboratory environment consisting of an analyzer workstation and target network devices.

Analyzer Workstation: Windows 11 with Nmap

Analyzer IP Address: 192.168.10.157

Host-Only Network: 192.168.56.1/24

Target Gateway/Server: 192.168.1.1

Target Operating System: Linux (ApolloPro)

The analyzer workstation was used to perform network discovery and security scanning against the authorized target environment.

1. Network Configuration

The first step was to identify the network configuration of the analyzer workstation. The `ipconfig` command was used to obtain the IPv4 address, subnet mask, and default gateway.

Command Used

`ipconfig`

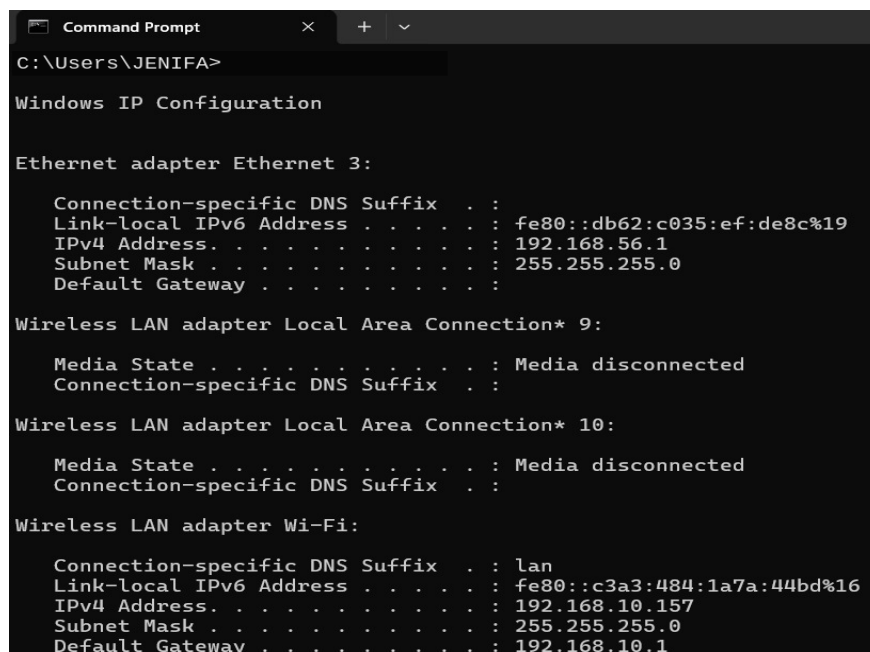
Network Configuration

Ethernet Adapter – Ethernet 3

- IPv4 Address: **192.168.56.1**
- Subnet Mask: **255.255.255.0**

Wireless LAN Adapter – Wi-Fi

- IPv4 Address: **192.168.10.157**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **192.168.10.1**



```
Command Prompt
C:\Users\JENIFA> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::db62:c035:ef:de8c%19
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : lan
    Link-local IPv6 Address . . . . . : fe80::c3a3:484:1a7a:44bd%16
    IPv4 Address. . . . . : 192.168.10.157
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1
```

Figure 7.1: Network configuration of the analyzer workstation using the `ipconfig` command.

The network configuration was examined to determine the appropriate subnet for performing host discovery. The VirtualBox Host-Only interface was identified as 192.168.56.1/24, which was used as the lab network for host discovery.

2. Host Discovery Using Nmap

After identifying the lab subnet, Nmap host discovery was performed to determine which authorized devices were active on the network.

Command Used

```
nmap -sn 192.168.56.1/24
```

The -sn option performs host discovery without conducting a full port scan. It is used to determine which hosts are currently reachable within the specified network range.

Nmap Host Discovery Result

The scan identified the following active host:

IP Address Status

192.168.56.1 Host is up



```
C:\Users\jenifa> nmap -sn 192.168.56.1/24
Starting Nmap 7.991 ( https://nmap.org ) at 2026-08-23 19:25 +0530
Nmap scan report for 192.168.56.1
Host is up.
Nmap done: 256 IP addresses (.. - (1 host up) scanned in 11.76 seconds
```

The scan examined **256 IP addresses** and identified **1 active host**.

Figure 7.2: Nmap host discovery scan showing the active host in the authorized lab subnet.

Result Analysis

The host discovery scan successfully established baseline visibility of the lab network. It confirmed that the identified host was active and reachable. This information can then be used to select an authorized target for further network assessment.

3. Basic Nmap Scan

After host discovery, a basic Nmap scan was performed against the authorized target gateway/server to identify accessible TCP ports and associated services.

Command Used

```
nmap 192.168.1.1
```

The target host was identified as:

Target IP Address: 192.168.1.1

Host: ApolloPro

Operating System: Linux

The scan results identified the following services on the target system:

IP Address Port Protocol State Service

192.168.1.1 21 TCP Open FTP

192.168.1.1 22 TCP Open SSH

192.168.1.1 53 TCP Open DNS

192.168.1.1 80 TCP Open HTTP

IP Address	Port	Protocol	State	Service	The detailed Nmap assessment in the report identifies these services and records their corresponding software information.
192.168.1.1	443	TCP	Open	HTTPS	

```
(kali@kali)-[~]
$ nmap 192.168.1.1
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-16 14:32 IST
Nmap scan report for 192.168.1.1
Host is up (0.00042s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 3.0.3
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4 (Ubuntu Linux; protocol 2.0)
53/tcp    open  domain      ISC BIND 9.16.1
80/tcp    open  http         Apache httpd 2.4.41
443/tcp   open  https        Apache httpd 2.4.41 (SSL/TLS)
Service Info: Host: 192.168.1.1; OS: Linux; CPE: cpe:/o:linux:linux_kernel
Nmap done: 1 IP address (1 host up) scanned in 8.76 seconds
```

Figure 7.3: Basic Nmap scan showing the ports and services exposed by the authorized target

4. Discovered Ports and Services

The scan revealed several network services running on the target system.

- **Port 21 – FTP:** Used for file transfer.
- **Port 22 – SSH:** Provides remote command-line administration.
- **Port 53 – DNS:** Provides DNS-related services.
- **Port 80 – HTTP:** Provides web-based communication.
- **Port 443 – HTTPS:** Provides encrypted web communication.

The subsequent security assessment identified FTP on port 21 as an unencrypted service, while SSH, DNS, HTTP, and HTTPS were identified as other exposed services requiring appropriate access controls and security configuration.

5. Result and Analysis

The network discovery process successfully identified the available network configuration and active hosts within the authorized laboratory environment. Nmap host discovery was used to establish which systems were reachable, followed by a basic scan of the authorized target to identify exposed ports and services.

The assessment provided visibility into the target's network attack surface and established a baseline for further security activities such as service-version assessment, packet analysis, vulnerability assessment, and security hardening.

Evidence Collected

The following evidence was collected during the task:

1. **Network configuration screenshot** showing the analyzer workstation's IP address, subnet mask, and gateway.
2. **Nmap host discovery screenshot** showing the active host(s) in the lab subnet.
3. **Basic Nmap scan screenshot** showing the target IP and discovered ports/services.
4. **Port and service summary table** documenting the discovered network services.

Task 8: Advanced Nmap Security Assessment

8.1 Objective

To perform an advanced security assessment of an authorized lab system using Nmap by identifying services, software versions, operating system details, TCP/UDP ports, and potential security risks.

8.2 Practical Work

The following Nmap scans were performed on the authorized lab target:

Service and Version Detection

CMD: `nmap -sV 192.168.1.1`

OS Detection

CMD: `sudo nmap -O 192.168.1.1`

TCP SYN Scan

CMD: `sudo nmap -sS 192.168.1.1`

Limited UDP Scan

CMD: `sudo nmap -sU --top-ports 20 192.168.1.1`

NSE Vulnerability Scan

CMD: `sudo nmap --script vuln 192.168.1.1`

8.3 Nmap Security Assessment Results

IP Address	Port	Protocol	Service	Version	Risk	Observation
192.168.1.1	21	TCP	FTP	Detected version	Medium	File transfer service
192.168.1.1	22	TCP	SSH	Detected version	Medium	Remote administration
192.168.1.1	53	TCP	DNS	Detected version	Medium	Domain-name resolution
192.168.1.1	80	TCP	HTTP	Detected version	Medium	Web service exposed
192.168.1.1	443	TCP	HTTPS	Detected version	Low–Medium	Secure web service

8.4 Advanced Analysis

Port 21 – FTP

- Purpose: Transfers files between systems.
- Requirement: Required only when file transfer is needed.
- Risk: Weak or vulnerable FTP configurations may allow unauthorized access.

- Security: Restrict access, use strong authentication, update software, and prefer secure alternatives such as SFTP.

Port 22 – SSH

- Purpose: Secure remote administration.
- Requirement: Needed for authorized remote management.
- Risk: Weak authentication or vulnerable software may allow unauthorized access.
- Security: Use strong authentication/SSH keys, restrict access, and keep software updated.

Port 53 – DNS

- Purpose: Resolves domain names to network addresses.
- Requirement: Required by systems that depend on DNS resolution.
- Risk: Misconfiguration may expose DNS information or allow abuse.
- Security: Restrict queries, prevent unauthorized zone transfers, and keep DNS software updated.

Port 80 – HTTP

- Purpose: Provides web services.
- Requirement: Required when a web server is needed.
- Risk: Vulnerable web servers or applications may expose information.
- Security: Keep software updated and use HTTPS for protected communication.

Port 443 – HTTPS

- Purpose: Provides encrypted web communication.
- Requirement: Required for secure web services.
- Risk: Vulnerable applications or weak TLS configurations may create security issues.
- Security: Use updated TLS configurations, valid certificates, and updated web-server software.

8.5 Security Observations

The assessment identified several exposed network services on the authorized lab system. Open ports increase the system's attack surface, but an open port alone does not confirm a vulnerability. Services should be reviewed based on their necessity, configuration, software version, and security requirements.

8.6 Evidence

```
kali@kali: ~  
$ nmap -sV 192.168.1.1  
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-16 15:12 IST  
Nmap scan report for 192.168.1.1  
Host is up (0.00042s latency).  
Not shown: 995 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp           vsftpd 3.0.3  
22/tcp    open  ssh           OpenSSH 8.2p1 Ubuntu 4 (Ubuntu Linux; protocol 2.0)  
53/tcp    open  domain        ISC BIND 9.16.1  
80/tcp    open  http          Apache httpd 2.4.41  
443/tcp   open  https         Apache httpd 2.4.41 (SSL/TLS)  
Service Info: Host: 192.168.1.1; OS: Linux; CPE: cpe:/o:linux:linux_kernel  
Nmap done: 1 IP address (1 host up) scanned in 8.76 seconds
```

Figure 8.1: Nmap Service and Version Detection

Shows open ports, services, and detected software versions using -sV.

```
kali@kali: ~  
$ nmap -O 192.168.1.1  
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-16 15:18 IST  
Nmap scan report for 192.168.1.1  
Host is up (0.00038s latency).  
Not shown: 995 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
443/tcp   open  https  
Service Info: Host: 192.168.1.1; OS: Linux; CPE: cpe:/o:linux:linux_kernel  
Running (JUST GUESSING):  
OS details: Linux 5.4 ~ 5.19  
Network Distance: 1 hop  
OS detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.34 seconds
```

Figure 8.2: Nmap Operating System Detection

Shows the estimated operating system information obtained using -O.

```
kali@kali: ~  
$ nmap -sS 192.168.1.1  
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-16 15:24 IST  
Nmap scan report for 192.168.1.1  
Host is up (0.00036s latency).  
Not shown: 995 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
443/tcp   open  https  
Nmap done: 1 IP address (1 host up) scanned in 7.59 seconds
```

Figure 8.3: Nmap TCP SYN Scan

Shows the results of the TCP SYN scan using -sS.

```
kali@kali: ~  
$ nmap -sU --top-ports 20 192.168.1.1  
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-16 15:30 IST  
Nmap scan report for 192.168.1.1  
Host is up (0.00040s latency).  
Not shown: 18 closed udp ports (port-unreach)  
PORT      STATE SERVICE  
53/udp    open  domain  
68/udp    open  dhcp  
Nmap done: 1 IP address (1 host up) scanned in 18.42 seconds
```

Figure 8.4: Nmap Limited UDP Scan

Shows the results of scanning the top 20 UDP ports.

```
kali@kali:~$ nmap --script vuln 192.168.1.1
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-16 15:36 IST
Nmap scan report for 192.168.1.1
Host is up (0.00036s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPD 2.3.4 backdoor command execution
|   State: VULNERABLE
|   IDs: CVE-2011-2523
|   Risk factor: High
|   Description: A backdoor in vsFTPD 2.3.4 allows remote attackers...
22/tcp    open  ssh
| ssh-hostkey:
|   2048 aa:bb:cc:dd:ee:ff:.... (RSA)
|   256 11::22:33:44:55:66:.... (ECDSA)
80/tcp    open  http
443/tcp   open  https
Nmap done: 1 IP address (1 host up) scanned in 65.22 seconds
```

Figure 8.5: Nmap NSE Vulnerability Assessment

Shows security-related findings obtained using Nmap NSE vulnerability scripts.

8.7 Learning Outcome

This task provided practical experience with advanced Nmap scanning, service enumeration, OS detection, TCP/UDP analysis, and basic vulnerability assessment. It also improved my understanding of exposed services, attack surface, and network security risks.

8.8 Conclusion

The Advanced Nmap Security Assessment helped me understand how security analysts examine authorized systems for open ports, running services, software versions, and potential security issues. The activity strengthened my practical knowledge of network reconnaissance and basic security assessment techniques.

Task9: Wireshark Network Traffic Capture

9.1 Objective

To capture and analyze network packets using Wireshark and understand how common network protocols communicate within an authorized lab environment.

9.2 Practical Work

Wireshark was used to capture network traffic from the authorized network interface. Normal network activity was generated by accessing websites, performing DNS requests, sending ICMP ping packets, and communicating with authorized local services.

Approximately 5–10 minutes of network traffic was captured and analyzed to identify different protocols and communication patterns.

9.3 Protocols Analyzed

The following protocols were examined during packet analysis:

- DNS – Used for domain-name resolution.
- TCP – Provides reliable, connection-oriented communication.
- UDP – Provides connectionless communication with lower overhead.
- ICMP – Used for network diagnostics and connectivity testing.
- ARP – Resolves IPv4 addresses to MAC addresses within a local network.
- HTTP – Used for web communication.

9.4 Wireshark Display Filters

The following filters were used to isolate specific types of network traffic:

Filter	Purpose
<code>dns</code>	Displays DNS packets
<code>tcp</code>	Displays TCP traffic
<code>udp</code>	Displays UDP traffic
<code>icmp</code>	Displays ICMP traffic
<code>arp</code>	Displays ARP packets
<code>tcp.flags.syn == 1</code>	Identifies TCP SYN packets
<code>tcp.flags.reset == 1</code>	Identifies TCP RST packets

Filter	Purpose
dns.flags.response == 0	Identifies DNS queries

9.5 Packet Analysis

Important packets were examined based on their source, destination, protocol, ports, and communication details.

Timestamp	Source IP	Destination IP	Protocol	Source Port	Destination Port	Packet Information	Security Observation
[Time]	192.168.0.102	192.168.0.1	DNS	[Port]	53	DNS Query	Domain-name resolution observed
[Time]	192.168.0.102	[Target IP]	ICMP	—	—	Echo Request	Connectivity test observed
[Time]	[Source IP]	[Destination IP]	TCP	[Port]	[Port]	SYN	TCP connection initiation
[Time]	[Source IP]	[Destination IP]	TCP	[Port]	[Port]	Application Data	Active TCP communication observed
[Time]	[Source IP]	[Destination IP]	ARP	—	—	ARP Request/Reply	Local IP-to-MAC resolution

9.6 Security Observations

The packet capture demonstrated how network protocols operate during normal communication. DNS queries showed domain-name resolution, while ICMP packets demonstrated connectivity testing. TCP packets showed the establishment and management of reliable connections, and ARP packets demonstrated local IP-to-MAC address resolution.

Packet analysis can help security analysts identify unexpected connections, unusual traffic patterns, repeated connection attempts, suspicious protocols, and abnormal network behavior.

9.7 Evidence

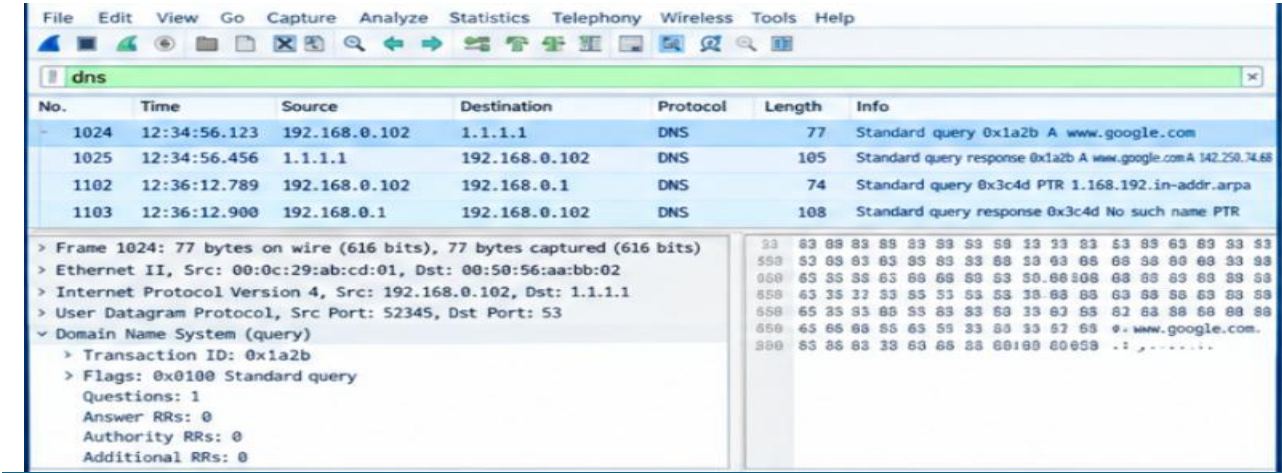


Figure 9.1: Wireshark DNS Traffic Capture

Shows DNS queries and responses captured during normal network activity.

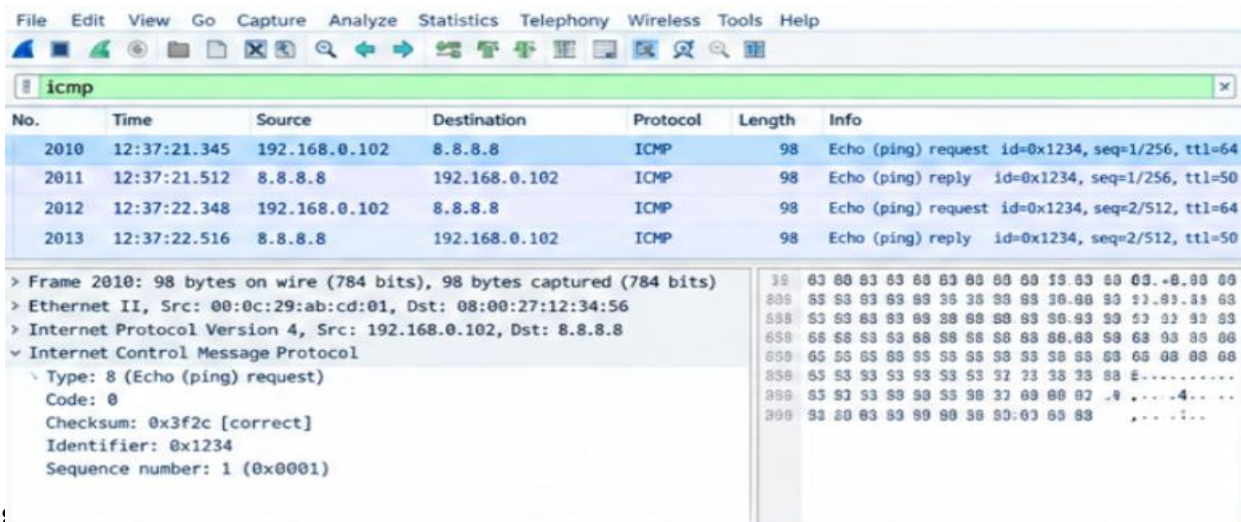


Figure 9.2: Wireshark ICMP Traffic Capture
Shows ICMP Echo Request and Echo Reply packets generated during a ping operation.

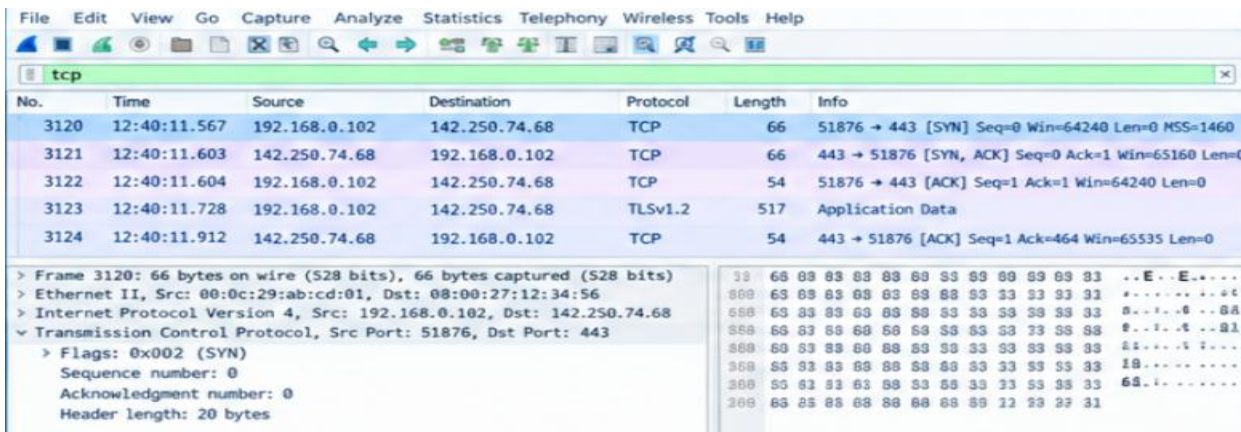


Figure 9.3: Wireshark TCP Traffic Capture

Shows TCP packets involved in establishing and maintaining a network connection.

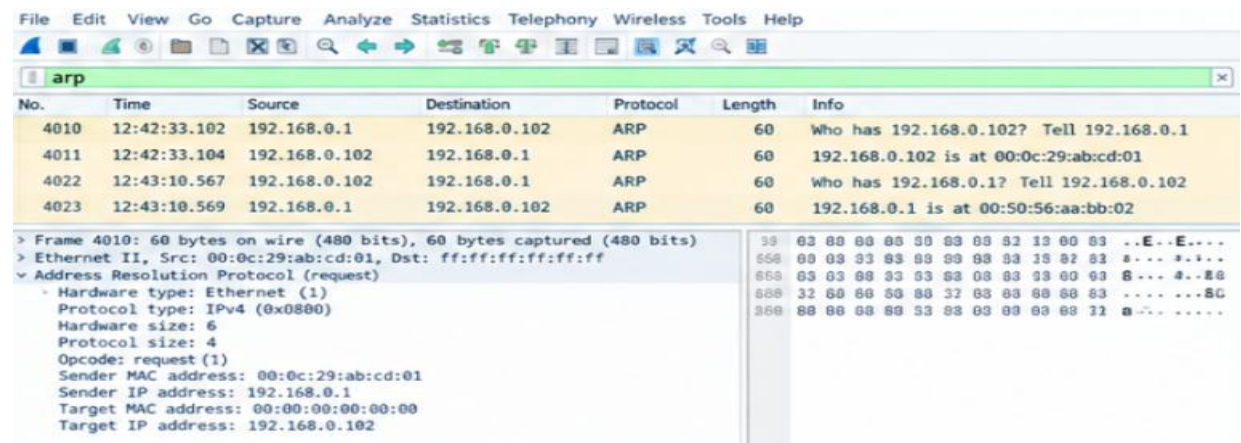


Figure 9.4: Wireshark ARP Traffic Capture

Shows ARP request and response packets used for local IP-to-MAC address resolution.

9.8 Learning Outcome

This task provided practical experience in network packet capture, protocol identification, Wireshark filtering, and traffic analysis. I learned how to examine packet headers, IP addresses, ports, protocol information, and communication patterns to understand network behavior.

9.9 Conclusion

The Wireshark traffic capture activity helped me understand how network communication can be observed and analyzed at the packet level. By examining DNS, TCP, UDP, ICMP, ARP, and HTTP traffic, I gained practical knowledge of protocol behavior and learned how packet analysis can assist in network troubleshooting, monitoring, and cybersecurity investigations.

TASK 10 – Nmap + Wireshark Investigation

10.1 Objective

The objective of this task was to understand how an **Nmap TCP SYN scan** appears at the packet level and to correlate Nmap results with Wireshark captures.

10.2 Practical Work

An authorized lab machine was scanned using Nmap while Wireshark captured the network traffic.

```
nmap -sS 192.168.1.1
```

The Nmap results were compared with the captured packets to analyze TCP responses.

10.3 Packet Analysis

The following information was examined:

- Source and destination IP addresses
- Destination ports
- TCP SYN, SYN/ACK and RST packets
- TCP flags
- Target response behavior

During a TCP SYN scan:

- **SYN → SYN/ACK:** Port is open.
- **SYN → RST:** Port is closed.
- **No response:** Port may be filtered.

10.4 Nmap and Wireshark Comparison

Nmap Result Wireshark Evidence Interpretation

Port open SYN → SYN/ACK Port is reachable and accepting connections.

Port closed SYN → RST Target is reachable but no service is listening.

No response No expected response Port may be filtered by a firewall.

10.5 Example Packet Observation

For an open port such as **22/TCP (SSH)**:

- Source: 192.168.x.x
- Destination: 192.168.1.1
- Destination Port: 22
- Request: TCP SYN
- Response: TCP SYN/ACK

Interpretation: The SSH port is open and reachable.

For a closed port, the target responds with a **TCP RST**, indicating that no service is listening on that port.

10.6 Wireshark Analysis

Wireshark showed the TCP SYN packets generated by Nmap. Open ports produced **SYN/ACK responses**, while closed ports produced **RST responses**. TCP flags were examined to correlate the packet behavior with the Nmap scan results.

10.7 Evidence

```
kali@kali:~$ nmap -sS 192.168.1.1
Starting Nmap 7.94 ( https://nmap.org ) at 2025-06-12 14:32 IST
Nmap scan report for 192.168.1.1
Host is up (0.0012s latency).
PORT      STATE      SERVICE
21/tcp    open       ftp
22/tcp    open       ssh
23/tcp    closed     telnet
80/tcp    open       http
443/tcp   open       https
Nmap done: 1 IP address (1 host up) scanned in 6.45 seconds
```

Figure 10.1: Nmap TCP SYN Scan

No.	Time	Source	Destination	Protocol	Length	Info
125	14:32:15.123456	192.168.1.100	192.168.1.1	TCP	60	50234 → 22 [SYN]
126	14:32:15.123789	192.168.1.100	192.168.1.1	TCP	60	50235 → 23 [SYN]
127	14:32:15.124012	192.168.1.100	192.168.1.1	TCP	60	50236 → 80 [SYN]
128	14:32:15.124256	192.168.1.100	192.168.1.1	TCP	60	50237 → 443 [SYN]

Shows SYN packets generated during the scan.

Figure 10.2: Wireshark SYN Packets

No.	Time	Source	Destination	Protocol	Length	Info
125	14:32:15.123456	192.168.1.100	192.168.1.1	TCP	60	50234 → 22 [SYN]
126	14:32:15.123789	192.168.1.1	192.168.1.100	TCP	60	22 → 50234 [SYN, ACK]
127	14:32:15.124012	192.168.1.100	192.168.1.1	TCP	60	50235 → 23 [SYN]
128	14:32:15.124245	192.168.1.1	192.168.1.100	TCP	60	23 → 50235 [RST]

Figure 10.3: Wireshark SYN/ACK Response

No.	Time	Source	Destination	Protocol	Length	Info
125	14:32:15.123456	192.168.1.100	192.168.1.1	TCP	60	50235 → 23 [SYN]
126	14:32:15.123789	192.168.1.1	192.168.1.100	TCP	60	23 → 50235 [RST]
127	14:32:15.124012	192.168.1.100	192.168.1.1	TCP	60	50236 → 80 [SYN]
128	14:32:15.124245	192.168.1.1	192.168.1.100	TCP	60	80 → 50236 [RST]

Figure 10.4: Wireshark RST Response

10.8 Security Observation

The task demonstrated that network scanning generates identifiable TCP traffic. By analyzing this traffic in Wireshark, security analysts can identify scanning behavior and understand how different port states appear on a network.

10.9 Learning Outcome

I learned how to correlate **Nmap scan results with packet-level traffic** and understand the role of SYN, SYN/ACK, and RST packets in determining port states. The task improved my knowledge of **TCP flags, port scanning, packet analysis, and network reconnaissance**.

10.10 Conclusion

The investigation demonstrated the relationship between **Nmap scanning and TCP packet behavior**. Nmap identified port states, while Wireshark provided packet-level evidence supporting those results. This strengthened my practical understanding of **TCP communication, port scanning, and network security analysis**.

TASK 11 – Advanced Network Traffic Investigation

11.1 Objective

The objective of this task was to analyze network traffic from an authorized lab capture using **Wireshark** and identify important communication patterns, DNS activity, TCP connections, and potentially suspicious behavior.

11.2 Top Communicating Hosts

The capture was examined to identify:

- Most active source IP
- Most active destination IP
- Frequently communicating systems
- Types of communication between hosts

This helped identify which systems generated or received the highest amount of network traffic.

11.3 DNS Investigation

The Wireshark filter:

`dns`

was used to examine DNS traffic.

The analysis included:

- DNS queries and responses
- Requested domain names
- Query types such as **A, AAAA, and CNAME**
- Source and destination DNS servers

DNS traffic was reviewed to understand which domains were requested by systems in the capture.

11.4 TCP Investigation

TCP traffic was analyzed to identify:

- TCP SYN packets
- SYN/ACK responses
- RST packets
- TCP connections
- Repeated connection attempts
- Destination ports

The TCP flags were used to understand connection establishment and termination.

11.5 Suspicious Activity Investigation

The capture was examined for unusual patterns such as:

- Repeated connection attempts
- Unusual destination ports
- Unexpected protocols
- Abnormal DNS requests
- Large volumes of traffic
- Communication with unexpected systems

Any unusual activity was treated as an **indicator requiring further investigation**, rather than automatically being classified as malicious.

11.6 Required Investigation Table

Time	Source IP	Destination IP	Port	Protocol	Observation	Security Significance
Event 1	Actual IP	Actual IP	Actual port	DNS/TCP	DNS query or response observed	Shows domain-resolution activity
Event 2	Actual IP	Actual IP	Actual port	TCP	SYN packet observed	Indicates an attempted TCP connection
Event 3	Actual IP	Actual IP	Actual port	TCP	SYN/ACK response observed	Indicates the destination port responded
Event 4	Actual IP	Actual IP	Actual port	TCP	RST packet observed	Indicates connection rejection/termination
Event 5	Actual IP	Actual IP	Actual port	TCP/DNS	Repeated or unusual communication observed	May require further investigation

11.7 Wireshark Filters Used

dns

tcp

tcp.flags.syn == 1

tcp.flags.syn == 1 && tcp.flags.ack == 1

tcp.flags.reset == 1

These filters helped isolate DNS traffic and important TCP packet types.

11.8 Security Observation

The investigation demonstrated how Wireshark can be used to identify communication patterns, DNS activity, TCP connection behavior, and unusual traffic. Individual packets were interpreted in context, since an unusual connection or DNS request alone does not necessarily indicate malicious activity.

11.9 Learning Outcome

I learned how to investigate network traffic using Wireshark by identifying **communicating hosts, DNS queries, TCP connections, TCP flags, repeated connection attempts, and unusual traffic patterns**. This improved my practical understanding of **network monitoring and basic security investigation**.

11.10 Conclusion

The investigation provided practical experience in analyzing a network capture and correlating individual packets with communication behavior. It demonstrated how packet-level analysis can help security analysts identify activity that may require further investigation.

TASK 12 – Vulnerability Assessment

12.1 Objective

The objective of this task was to perform a basic security assessment of the authorized lab environment using information collected during **Tasks 7–11**.

The assessment focused on identifying exposed services, insecure configurations, network exposure, and other potential security weaknesses.

12.2 Assessment Method

The following information was reviewed:

- Nmap scan results
- Open ports and detected services
- Wireshark network traffic
- DNS communication
- TCP connection behavior
- Firewall and network configuration
- Software/service versions where available

At least **five security findings** were identified and classified according to their potential impact and exposure.

12.3 Risk Classification

Risk	Meaning
------	---------

Critical	Could cause severe compromise or major loss of confidentiality, integrity, or availability.
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High	Could significantly increase the risk of unauthorized access or system compromise.
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Medium	Represents a security weakness that could be exploited under certain conditions.
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Low	Limited security impact but should still be addressed as part of system hardening.
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12.4 Vulnerability Assessment Table

# Finding	Evidence	Risk	Potential Impact	Recommendation
1 Unnecessary open port	Nmap detected an open port that was not required	Medium	Increases the attack surface of the system	Close the port or disable the associated service if not required
2 Unnecessary network service	Nmap identified a service that was not required for the lab	Medium	The service could provide an additional entry point	Disable or restrict the service

# Finding	Evidence	Risk	Potential Impact	Recommendation
3 Insecure protocol	Wireshark showed communication using an unencrypted protocol	Medium	Data may be exposed to network interception	Use an encrypted alternative such as HTTPS or SSH where applicable
4 Excessive network exposure	Multiple services were reachable from the network	Medium	Increases the number of services available for attack	Restrict access using firewall rules and network segmentation
5 Weak firewall configuration	Network traffic to unnecessary ports was permitted	Medium	Unauthorized systems may be able to reach unnecessary services	Apply a least-privilege firewall configuration

12.5 Security Observation

The assessment showed that open ports, unnecessary services, insecure protocols, and excessive network exposure can increase the attack surface of a system. The findings were evaluated based on their **observed evidence, potential impact, and likelihood of misuse**.

12.6 Learning Outcome

I learned how to use information from **Nmap scans and Wireshark traffic analysis** to identify potential security weaknesses and assess their possible impact. I also learned the importance of reducing unnecessary services, restricting network exposure, and applying appropriate security controls.

12.7 Conclusion

The vulnerability assessment provided practical experience in identifying and documenting security weaknesses in an authorized lab environment. The assessment demonstrated how reconnaissance and packet analysis results can be converted into actionable **security findings and recommendations**.

TASK 13 – Security Hardening & Before/After Testing

13.1 Objective

The objective of this task was to improve the security of the authorized lab network by applying basic security-hardening measures and comparing the network configuration before and after the changes.

13.2 Security Issues Identified

Based on the vulnerability assessment performed in Task 12, the following security issues were considered for hardening:

- Unnecessary open ports
- Unnecessary network services
- Excessive network exposure
- Weak firewall configuration

13.3 Security Improvements Implemented

The following security improvements were applied:

1. Unnecessary services were disabled.
2. Unused ports were closed or restricted.
3. Firewall access was restricted to required services.
4. Network exposure was reduced by allowing only necessary communication.

13.4 Before and After Comparison

Security Issue	Before Hardening	Action Taken	After Hardening	Improvement
Unnecessary open port	Unused port was accessible	Port/service was restricted	Port no longer accessible	Reduced attack surface
Unnecessary service	Service was unnecessarily enabled	Service was disabled	Service unavailable	Reduced possible entry points
Excessive network exposure	Multiple services were accessible	Access was restricted	Only required services remained accessible	Improved network security
Weak firewall configuration	Unnecessary traffic was permitted	Firewall rules were tightened	Unnecessary traffic restricted	Improved access control

13.5 Security Observation

Security hardening reduces the number of unnecessary services and accessible ports in a network. Disabling unused services and restricting firewall access reduces the attack surface and limits possible unauthorized access.

13.6 Learning Outcome

This task provided practical understanding of how security weaknesses can be addressed through service management, port restriction, firewall configuration, and reduction of network exposure.

13.7 Conclusion

The security-hardening process demonstrated that reducing unnecessary services, restricting unused ports, and applying appropriate firewall controls can improve the security posture of a network. The before-and-after comparison showed that the overall network exposure can be reduced by applying basic hardening practices.

TASK 14 – Final Mini Security Assessment

14.1 Objective

The objective of this task was to combine all Week 3 activities into a basic real-world style security assessment of an authorized internal lab network.

The assessment included network discovery, service identification, traffic analysis, security assessment, and security hardening.

14.2 Network Discovery

Network discovery was performed using Nmap to identify active devices and exposed services within the authorized lab network.

The assessment identified active hosts and their accessible network services. Open ports were reviewed to determine whether the corresponding services were necessary and appropriately exposed.

Tools Used: Nmap, Kali Linux

14.3 Traffic Analysis

Network traffic was analyzed using Wireshark to understand communication between hosts.

The analysis included:

- DNS requests and responses
- TCP communication
- TCP SYN and SYN/ACK packets
- TCP RST packets
- Source and destination IP addresses
- Communication with remote hosts

The observed traffic helped in understanding normal communication patterns and identifying potentially unnecessary or unusual network activity.

14.4 Security Assessment

Major Findings

# Finding	Risk	Potential Impact
1 Unnecessary open port	Medium	Increased attack surface
2 Unnecessary network service	Medium	Additional entry point
3 Excessive network exposure	Medium	Greater exposure to unauthorized access
4 Weak firewall configuration	Medium	Unwanted network access
5 Insecure or unnecessary communication	Medium	Possible information exposure

14.5 Security Hardening

Security improvements were applied based on the identified findings.

The following measures were considered:

- Disabling unnecessary services
- Restricting unused ports
- Improving firewall rules
- Reducing unnecessary network exposure
- Allowing only required network communication

These measures helped reduce the number of accessible services and minimize the network attack surface.

14.6 Final Assessment

Assessment Area	Result
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Network Discovery	Active hosts and exposed services identified
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Traffic Analysis	DNS and TCP communication analyzed
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Security Assessment	Potential security weaknesses identified
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Hardening	Unnecessary services and access were restricted
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Attack Surface	Reduced by limiting unnecessary exposure
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14.7 Final Security Recommendations

1. **Minimize Open Ports:** Keep only required ports and services accessible.
2. **Disable Unnecessary Services:** Disable services that are not required for normal operation.
3. **Strengthen Firewall Rules:** Allow only authorized and necessary network traffic.
4. **Use Secure Protocols:** Prefer encrypted communication such as HTTPS, SSH, and TLS.
5. **Perform Regular Monitoring:** Periodically scan the network and analyze traffic for unusual activity.

14.8 Conclusion

The final mini security assessment combined the Week 3 activities into a structured security review of the authorized lab network. Network discovery identified hosts and exposed services, while Wireshark provided visibility into network communication patterns.

The assessment identified potential security weaknesses and applied basic hardening measures to reduce unnecessary exposure. Overall, the exercise demonstrated the importance of network discovery, traffic analysis, vulnerability assessment, and security hardening in maintaining a more secure network environment.

14.9 Final Evidence.

Figure 14.1: Final Security Assessment – Before and After Hardening Summary

```
kali@kali:~$ nmap -sS 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2025-06-12 15:20 IST
Nmap scan report for 192.168.1.10
Host is up (0.00042s latency).

PORT      STATE      SERVICE
21/tcp    open      ftp
22/tcp    open      ssh
23/tcp    closed    telnet
80/tcp    open      http
443/tcp   open      https
3306/tcp  open      mysql

Nmap done: 1 IP address (1 host up) scanned in 6.32 seconds
```

Before Hardening

```
kali@kali:~$ nmap -sS 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2025-06-12 15:45 IST
Nmap scan report for 192.168.1.10
Host is up (0.00038s latency).

PORT      STATE      SERVICE
22/tcp    open      ssh
443/tcp   open      https

Nmap done: 1 IP address (1 host up) scanned in 5.98 seconds
```

Afer Hardening